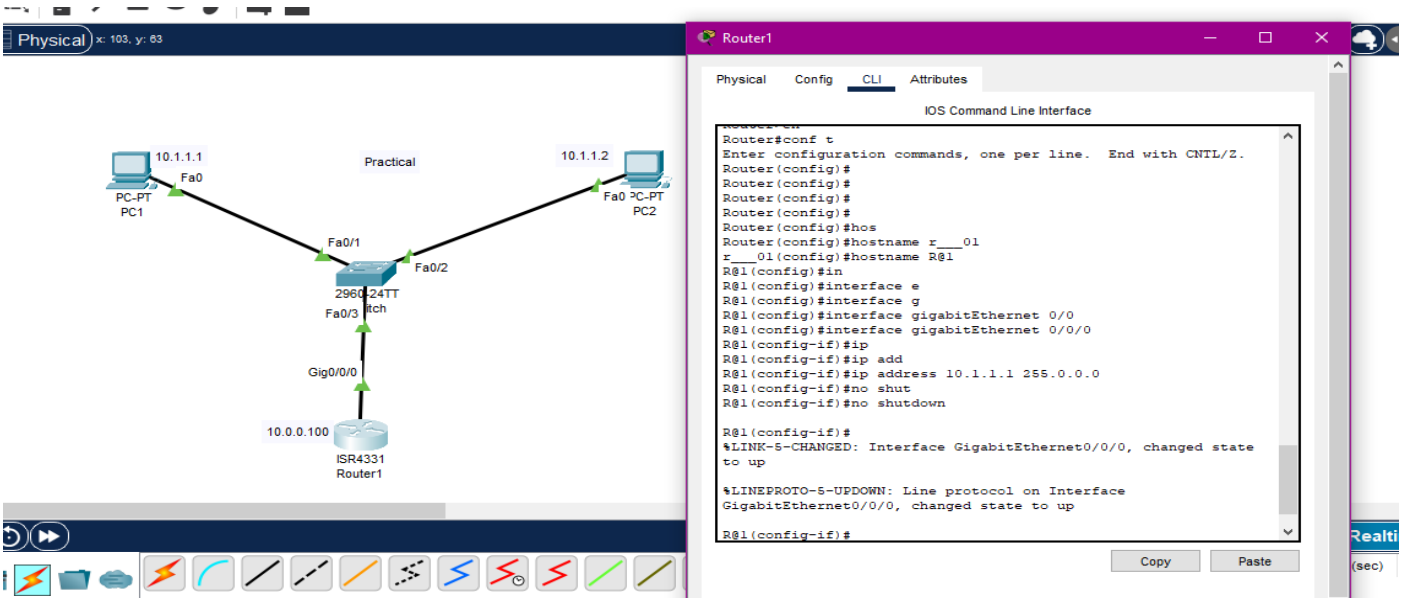
## WEEK 1: Networking Fundamentals | Days: 1-7

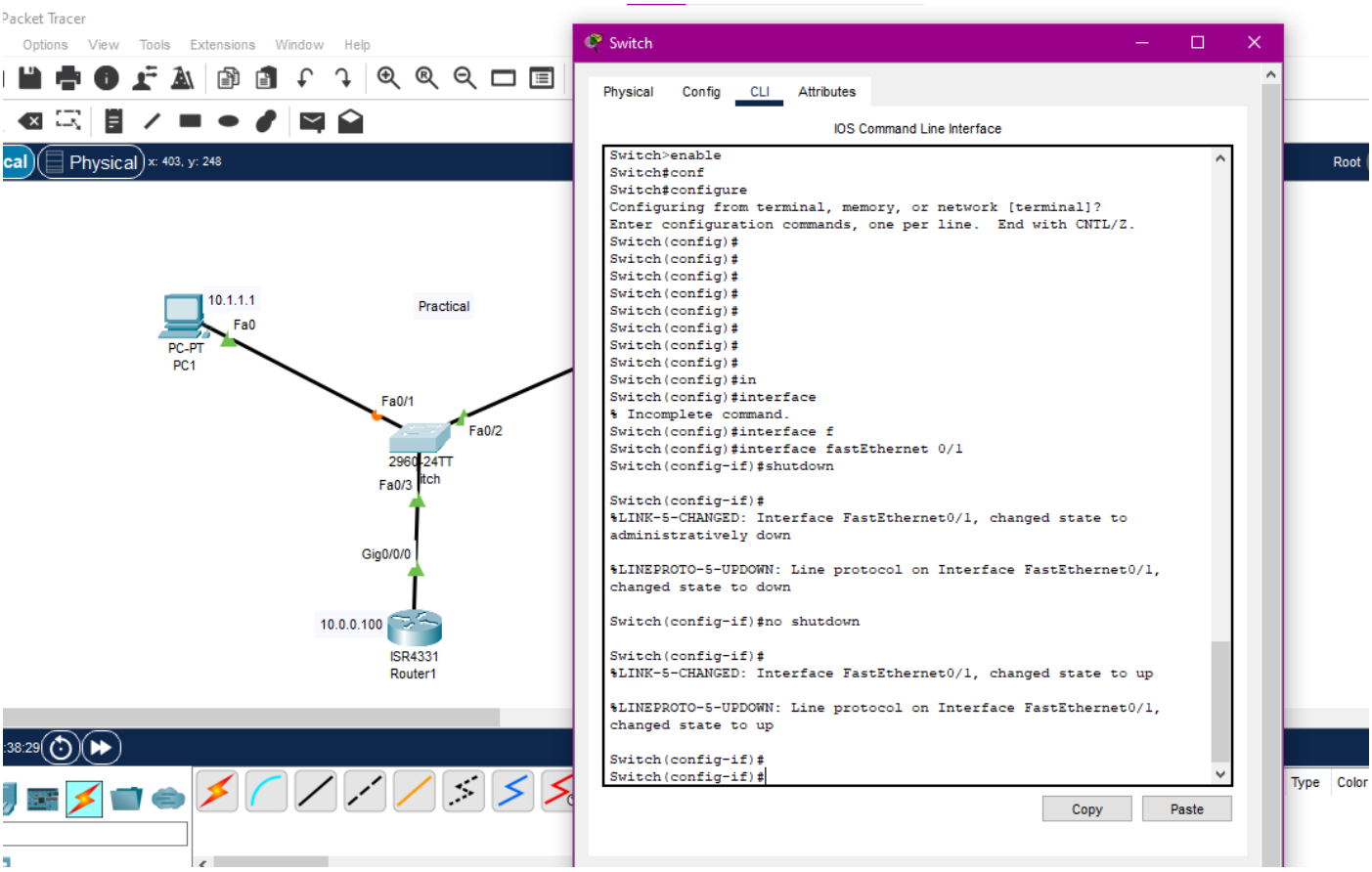
## Day 7 Tasks—Router and Switch Configuration

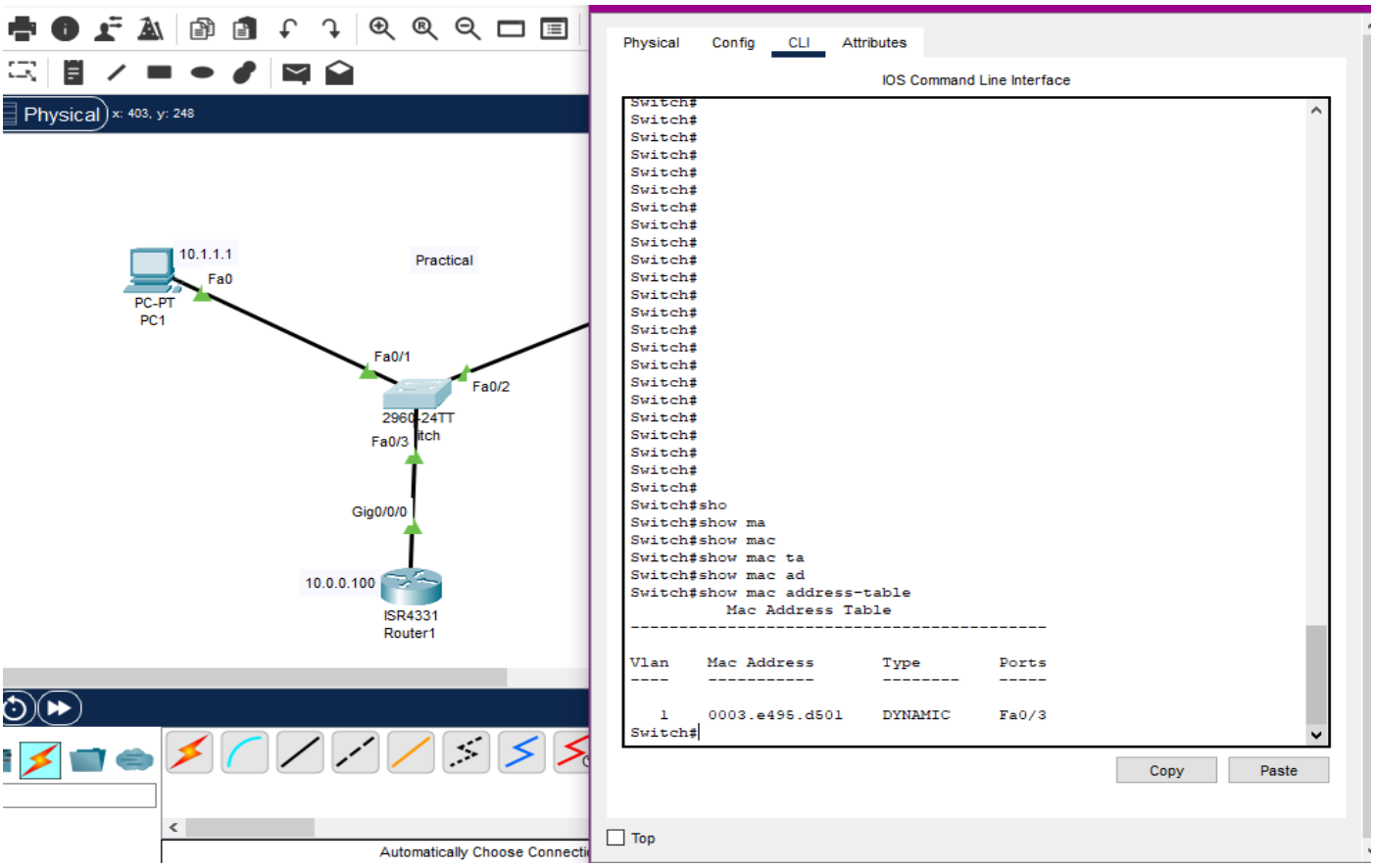
# Practical:

- **Configure router IP address via CLI.**




- **Enable and disable switch ports; verify the MAC address table.**






## ➢ TASK: Role of a router in a network.

A router is a networking device that forwards data packets between different computer networks. It connects multiple packet-switched networks or subnetworks, managing traffic by directing packets to their intended IP addresses. Routers allow multiple devices to share an Internet connection efficiently.

### How Does a Router Work?

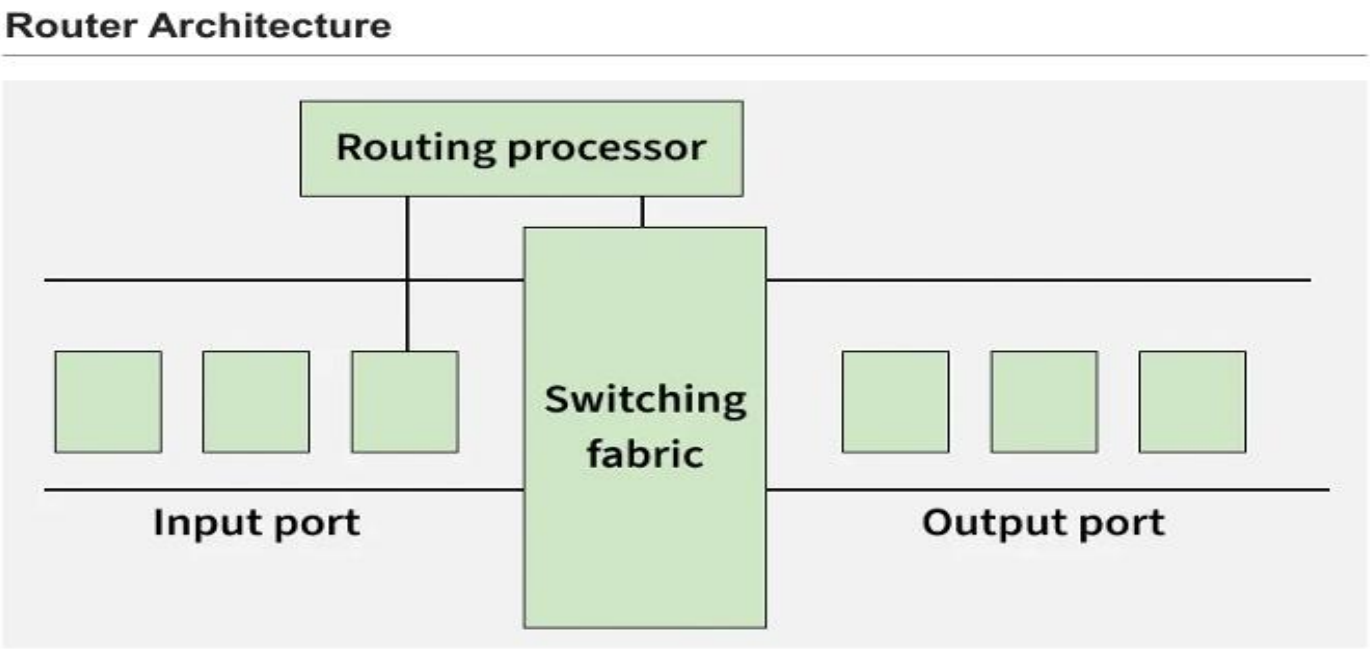
Routers determine the path for a packet by examining its destination IP address and consulting the routing table, which contains information on network paths. They use a set of rules to identify the most efficient route for each packet.

- **Static routing:** Configured manually, suitable for small or stable networks.
- **Dynamic routing:** Automatically updated based on network activity, ideal for large or changing networks.

## Functions of a Router

- **Forwarding:** Receives packets, examines headers, and forwards them to the correct output port.
- **Routing:** Determines the optimal path for packets using routing tables and algorithms.
- **Network Address Translation (NAT):** Translates private IPs to a public IP for Internet access.

- **Security: Supports** firewalls and other security measures.
- **VPN Connectivity:** Provides secure remote access to networks.
- **Bandwidth Management:** Controls data flow to prevent congestion.
- **Monitoring & Diagnostics:** Tracks traffic and helps troubleshoot network issues.




Architecture of Router

## A typical router consists of:

- **Input Port**: Accepts packets, decapsulates them, and determines forwarding paths.
- **Switching Fabric:** The core of the router connecting input ports to output ports.

## Can be implemented via:

- **Memory switching:** CPU copies packets to output ports.
- **Bus switching:** Single bus transfers packets to the correct port.
- **Interconnection networks:** Complex designs connecting multiple input/output ports.

**Output Port**: Transmits packets to outgoing links, managing queuing and link-layer functions.

**Routing Processor:** Executes routing protocols and algorithms, maintaining the forwarding table.

Common Routing Protocols

**Open Shortest Path First (OSPF):** Determines the optimal path across networks. Border Gateway Protocol **(BGP):** Shares routing information between edge routers.

**Interior Gateway Routing Protocol (IGRP):** Exchanges routing info within autonomous networks.

**Enhanced IGRP (EIGRP):** Requests routing paths from neighbors if unknown.

**Exterior Gateway Protocol (EGP):** Shares routing data between internet hosts.

## Applications of Routers

- Connect remote servers, networks, and devices globally.
- Support wired and wireless communication, including high-speed data transfer.
- Used by ISPs to transmit audio, video, image, and email efficiently.
- Implement access control, enabling selective resource usage.

## Types of Routers

- **Broadband Routers:** Connect computers to the Internet and share the connection.
- **Wireless Routers:** Create Wi-Fi networks in homes or offices.
- **Wired Routers:** Connect multiple devices via Ethernet cables, common in schools and offices.
- **Edge Routers:** Located at network boundaries, distributing packets to and from ISPs.
- **Core Routers:** Operate within networks, handling heavy data traffic.
- **Virtual Routers**: Software-based routers implemented on virtual machines for flexibility and scalability.
- **Portable Routers:** Small devices creating private Wi-Fi for mobility.

## Security Challenges in Routers

- **Vulnerability Exploits:** Firmware flaws can be exploited by attackers; regular updates are necessary.
- **DDoS Attacks:** Distributed Denial-of-Service attacks can overload routers.
- **Default Admin Credentials:** Weak or unchanged credentials can allow unauthorized access.

### How switches learn and use MAC address tables.

The MAC (Media Access Control) address is a 48-bit physical identity of a device used to uniquely identify a device in layer 2 of the OSI model. It is represented in hexadecimal. The 3 bytes of the address represent the OUI, and the last 3 bytes are unique to the device itself. The switch keeps track of the other Ethernet interfaces on the network to which it is linked in the MAC address table. Instead of broadcasting the data over all ports, the table allows the switch to send outgoing data (Ethernet frames) on the exact port needed to reach its destination (flooding). Two different types of entries are possible in the MAC address table:

- **Static:** The switch administrator must manually add static items to the table. Dynamic entries are less important than static entries. Static entries are active until the switch administrator deletes them.
- **Dynamic:** Through a process known as MAC learning, in which the switch retrieves the source MAC address (and VLAN ID, if present) of each Ethernet packet received on a port, dynamic entries are

automatically added to the table. If the table does not already include the obtained address, it is added. Dynamic entries are automatically erased after being present in the table for a certain period of time (specified by the command mac-address-table age-time).

The switch stores the MAC information in a table called the CAM table or the MAC table.

## CAM Table Stores:

- MAC address
- The interface
- VLAN MAC address belongs to

### MAC address table for Cisco switches:

Consider a topology given below: Take 3 PC and connected them with a Cisco Switch and then connect the Switch to a router at the end such as given below:

Router0 configuration: Go to Router0 and do the following configuration.

Interface GigabitEthernet0/0/0

ip address 10.1.1.1 255.0.0.0

no shutdown

**(The End)**